The Paleomagnetism of the Ring Complexes at Marangudzi and the Mateke Hills

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Abstract. Oriented rock cores have been drilled at nineteen sites in four intrusive ring complexes in the Nuanetsi igneous province of southeastern Southern Rhodesia. At Marangudzi eleven sites were distributed among nine rock types; the aim was to investigate the relation between rock type and remanent magnetization. Six of these sites, and one from another complex, gave no paleomagnetic results. At ten sites stable components of remanent magnetization, which remained after partial demagnetization in alternating fields, gave closely grouped directions approximately reversed with respect to the present geomagnetic field. These directions are considered to be those of thermoremanent magnetizations acquired when the rocks were intruded. The mean north paleomagnetic pole position from these ten sites is 60.9°S, 86.2°E, and $A_{95} = 6.5°$. Samples from Marangudzi have been dated from K-Ar decay at close to 190 m.y., and it is probable that the other complexes are of approximately the same age. Two sites in different complexes are normally magnetized, which suggests that the geomagnetic field changed sense from reversed to normal near the close of the intrusive episode. Evidence is presented which shows that relative movement between Africa and the pole was small during the Mesozoic era.

Geology of the ring complexes. Paleomagnetic results from four ring complexes in the southeast of Southern Rhodesia are discussed (Figure 1). These ring complexes form part of the Nuanetsi igneous province, which lies at the intersection of the Limpopo lineament with the volcanic monocline of the Lebombo [*Cox*, 1962]. The Lebombo lavas form part of the widespread series of volcanic rocks which were extruded late in the formation of the Karroo system. Of the complexes considered in this paper, three (*B*, *C*, and *D* of Figure 1) cut the Karroo lavas where Cox estimates their thickness at about 7600 m. The Marangudzi complex is intrusive into basement granite.

The geology of the Marangudzi complex (*A*, Figure 1) has been described by *Gifford* [1961] and by *Rees* [1960]. A roughly rectangular mass of gabbro is intersected by an oval ring of granite rocks. Within the granite ring the gabbro is intruded by syenites and foyaites which form a group of hills. The whole complex is about 11 km in diameter. The Nuanetsi igneous province has been studied by *Cox et al.* [1964]; an abstract of their memoir on the province is given by *Cox* [1962]. The Dembe-Divula complex (*B*, Figure 1) consists of a roughly circular mass, 8 km across, of granitic and granophyric rocks, with an extension of similar acid rocks to the northeast. The southwest rim of the intrusion is breached by a subsequent gabbro intrusion of irregular shape. The Masukwe complex (*C*) has a core of granophyre surrounded by a ring of microgabbro. The gabbroic ring is intruded by numerous microgranite dikes, which follow the curvature of the ring. The whole intrusion is roughly circular and about 8 km across. The northern ring complex (*D*) is generally basic in composition. Quartz gabbro, norite, and olivine gabbro form alternate annular bands of oval shape. In places the basic rocks are intruded by microgranite dikes which follow the general ring structure. The basic rocks are bounded by granophyre to the south and by metamorphic belts to the east and west. The gabbroic intrusions form an oval outline 11 km across from north to south and 8 km from east to west.

Aims of the investigation. The primary aim was to determine a paleomagnetic pole position

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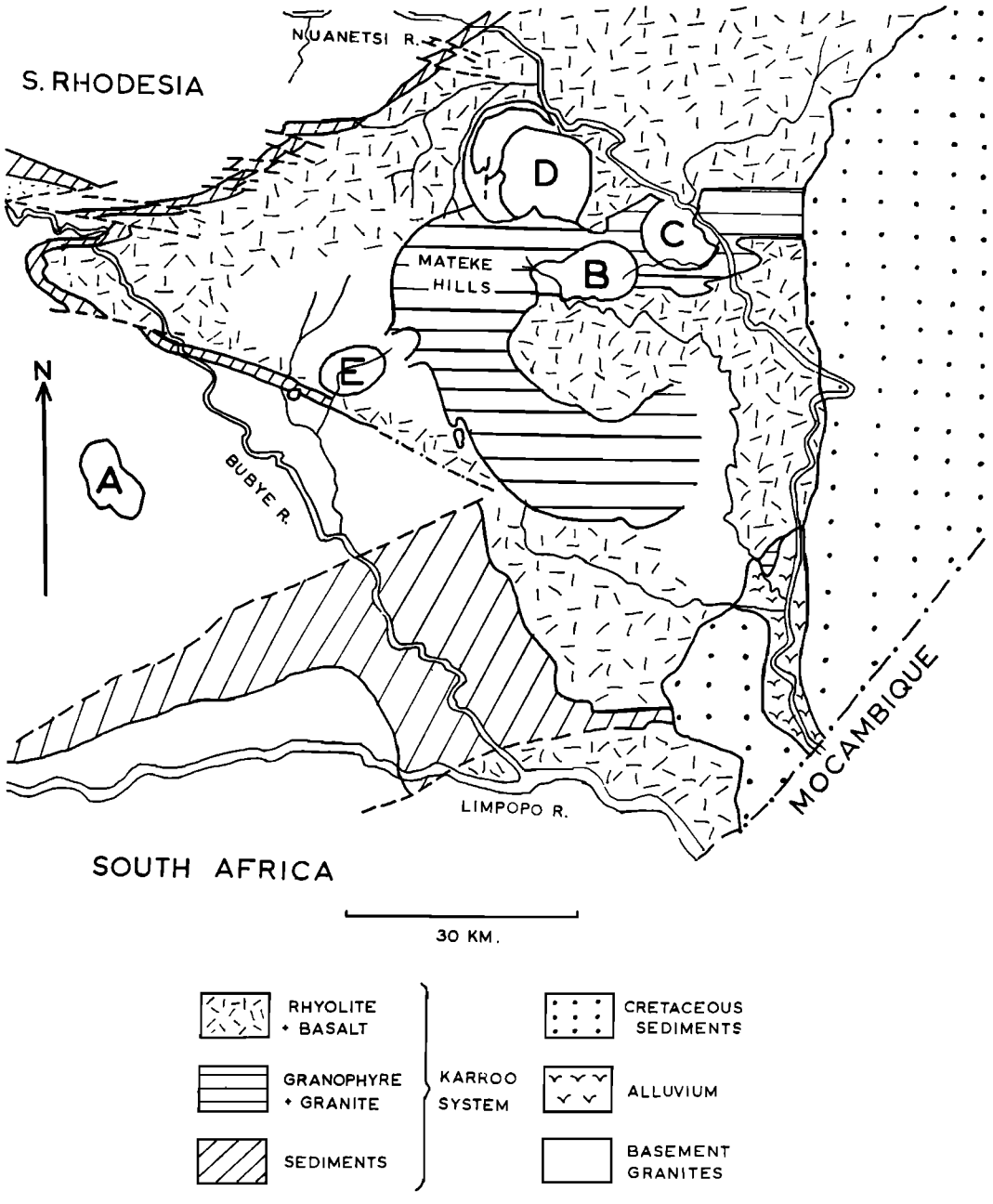


Fig. 1. Geology of the Nuanetsi igneous province (after C. J. Stillman). *A* is the Marangudzi ring complex, *B* the Dembe-Divula complex, *C* the Masukwe complex, and *D* the Northern ring complex. *E* is a granite intrusion.

as part of a general study of polar wander and continental drift [*Gough et al.*, 1964]. The initial investigation was limited to the Marangudzi complex; the wide variety of igneous rocks in that complex suggested the secondary aim, which was the sampling of nine different rock types in order to discover which rocks might have stable magnetizations which might represent the direction of thermoremanent magnetization. In this we were greatly helped by the

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detailed knowledge and advice of Dr. A. C. Gifford.

Field work. Two of us (A.B. and D.I.G.) visited the Marangudzi complex with Dr. Gifford in February 1962 and drilled oriented cores at eleven sites. At four sites (A1, A3, A7, and A9, Table 1) oriented cores 3.5 cm in diameter and about 3 m long were secured by means of the Gough-Opdyke drill [*Gough and Opdyke*, 1963]. About ten specimens were cut from each such core, and the mean direction of magnetization for these was used as a site mean direction. At the other seven sites a Graham-Keiller sampling drill [*Graham and Keiller*, 1960] was used to obtain about ten cores 2.5 cm in diameter and about 15 cm in length. Generally, two specimens were cut from each core. The individual cores were spaced 2 m or more apart, to randomize lightning effects. Orientation, relative to the sun's azimuth, was obtained by the technique described by *Gough and Opdyke* [1963]. The locations of the sites and rock types are given in Table 1.

It will be seen that useful paleomagnetic results were secured from five sites at Marangudzi. It was desirable to extend the study to the Mateke Hills intrusions (*B*, *C*, and *D*, Figure 1). Three of us (D.I.G., D.L.J., and N.D.O.) sampled gabbro at eight sites in these three complexes during May and June 1963. Dr. Gifford kindly gave his time and help in the selection of the sites. One site was sampled by means of the Gough-Opdyke drill (B2, Table 2); at the remainder the Graham-Keiller drill was used. Longer cores (18 to 53 cm) were drilled with the latter drill than at Marangudzi. The positions of the sites are given in Table 2.

TABLE 1. Sites Sampled in the Marangudzi Complex

Site No.	Lat. S	Long. E	Rock Type
A1	22°00′	30°39′	Green quartz syenite
A2	22°00′	30°39′	Small meandering dolerite dikes
A3	22°01′	30°39′	Pulaskite (nepheline syenite)
A4	22°02′	30°40′	Pseudoleucite phonolite dike
A5	22°03′	30°38′	Foyaite
A6	22°03′	30°39′	Gabbro
A7	22°04′	30°41′	Gabbro
A8	22°05′	30°40′	Microgabbro
A9	22°05′	30°39′	Granite
A10	22°05′	30°41′	Gabbro
A11	22°05′	30°40′	Lamprophyric dike

TABLE 2. Sites Sampled in Three Complexes in the Mateke Hills

Complex	Site No.	Lat. S	Long. E
Dembe-Divula	B1	21°52′	31°07′
	B2	21°52′	31°07′
Masukwe	C1	21°46′	31°15′
	C2	21°46′	31°16′
	C3	21°44′	31°18′
Northern ring	D1	21°40′	31°08′
	D2	21°40′	31°08′
	D3	21°40′	31°07′

Methods of measurement. Directions and intensities of remanent magnetization were measured by means of the spinner magnetometer described by *Gough* [1964]. The samples were partially demagnetized with apparatus described by *McElhinny and Gough* [1963].

Paleomagnetic results, Marangudzi ring complex. At four sites (A3, A7, A10, and A11, Table 1) the directions of magnetization of specimens from a site became closely grouped upon partial demagnetization in alternating fields, and the site mean directions were in good agreement. These four sites are believed to give the direction of TRM acquired at the date of intrusion and are discussed below. Two of these sites were in gabbro, one in a lamprophyric dike, and one in pulaskite. The rock types which gave stable and consistent magnetizations were, therefore, basic at three sites and undersaturated syenitic at the fourth. A fifth site (A8, microgabbro) gave well-grouped directions nearly opposite to those given by the other four.

Two sites (A2, 6) showed widely scattered directions of NRM, and the grouping was not significantly improved by alternating magnetic fields up to 800 gauss (the maximum field available). Two sites (A4, 5) showed scattered NRM directions which were little affected by alternating fields up to about 200 gauss. When specimens were demagnetized in larger fields, the directions of magnetization showed large, erratic changes. An extreme case of this type of behavior is illustrated in Figure 4*b*, which refers to site A9. The remanent magnetization after treatment is probably near the minimum pos-

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sible resultant for the population of domains present, when fully randomized [*Irving et al.*, 1961; *McElhinny and Gough*, 1963]. In Figure 2 the directions of NRM for nine cores at site A4 are shown, together with the directions of magnetization after treatment in an alternating field of maximum amplitude 200 gauss. In larger fields the directions undergo large erratic changes.

The intensities of magnetization of specimens from site A4, after treatment in a 200-gauss field, were between 1 and 5 per cent of the NRM intensities. Well-grouped stable directions of magnetization have been secured several times from specimens which retain much less than 1 per cent of the NRM intensity after alternating-field treatment [*McElhinny and Opdyke*, 1964]. There appears, therefore, to be no special significance in this ratio. A specimen may have a given intensity of NRM either if it has a high magnetic mineral content magnetized in a low field or if it has low magnetic mineral content magnetized in a high field. It is probable that instability from randomization of domains would be found at a higher remanent intensity in the former case. It is doubtful whether either the shape of the demagnetization curve or the ratio of final to initial intensity has much relevance to the choice of demagnetization field or to the validity of results. The real criterion is the manner of variation of the direction of magnetization with increasing field [*As and Zijderveld*, 1958; *Irving et al.*, 1961; *McElhinny and Gough*, 1963].

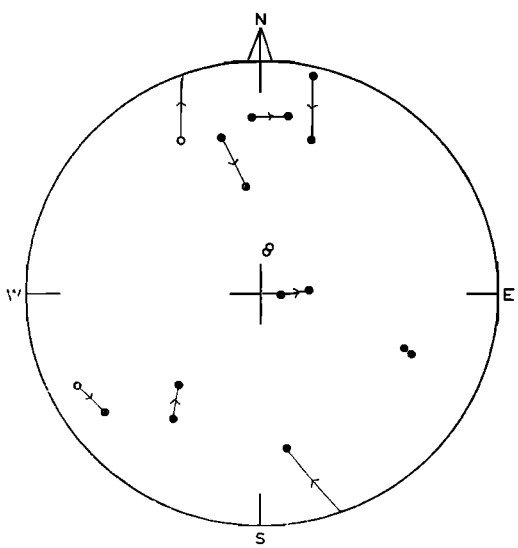


Fig. 2. Demagnetization of cores from site A4. Lower-hemisphere stereographic projection. Directions changed as indicated by arrows. Solid circles, north-seeking pole down; open circles, south-seeking pole down.

The irrelevance of the ratio of final intensity to initial (NRM) intensity is illustrated by Figure 3. This gives the directions of NRM of twelve specimens cut from the core drilled at site A7 (gabbro) with the Gough-Opdyke drill and the directions of magnetization after treatment in an alternating field of 300-gauss amplitude. Clearly no single event has produced the rather large disturbances in the NRM directions, but at least one lightning current would appear to be involved. The ratio of the final to the initial intensity varies from 0.01 for specimens near ground level to 0.5 for specimens 4 m below. The final intensities for the twelve specimens are all within a factor 2 of one another. Clearly, the lightning effects have produced intense isothermal magnetization near the surface and weaker effects lower down.

Figure 2 illustrates a typical distribution of directions of NRM found in outcrops of igneous rocks in central Africa, both for sites which give closely grouped directions on partial demagnetization in alternating fields and for sites like that illustrated, which do not. Another example is given by *McElhinny and Gough* [1963, Figure 3]. Each direction shown refers to a short core drilled with the Graham-Keiller drill. Within each core specimens give directions of NRM in close agreement, but there is wide disagreement between cores 2 m or more apart. The only cause known to us, of magnetic fields which differ widely in direction between points 2 or 3 m apart, is one or more electric currents at the same order of distance from these points. The roughly random distribution of directions of NRM therefore in itself strongly suggests lightning as the cause. Other arguments in support of this view are given by *McElhinny and Opdyke* [1964]. Detailed studies of lightning effects at an outcrop are given by *Graham* [1961] and *Cox* [1961].

The two rock types sampled at Marangudzi which have excess free quartz (oversaturated) have rather similar magnetic properties. Long cores were drilled from each with the Gough-Opdyke drill. Specimens from the green quartz

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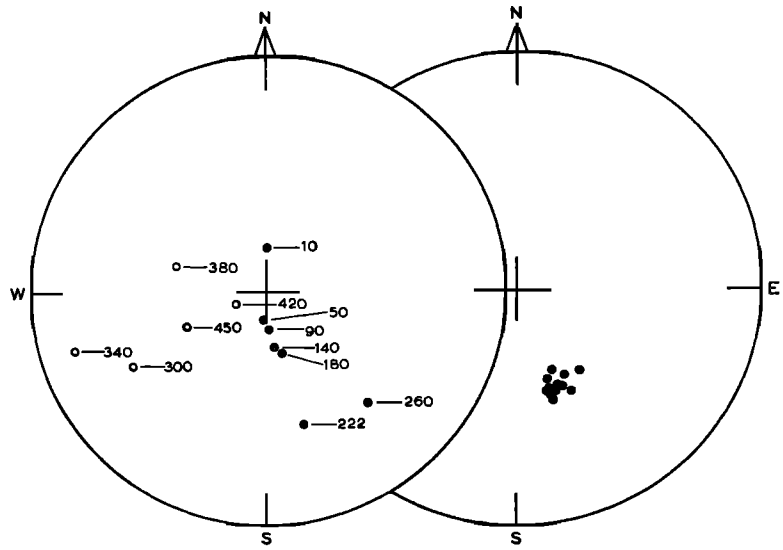


Fig. 3. Demagnetization of specimens from gabbro core, site A7. Left, NRM. Right, after treatment at 300 gauss. Numbers give depth in centimeters. Plotting convention as in Figure 2.

syenite core (site A1) gave directions of NRM which, with one exception, were well grouped and only 40° from the stable direction found at other sites. The directions of magnetization became unstable in fields less than 125 gauss, however, showing large erratic jumps as the field was increased. Specimens from the granite core (site A9) gave directions of NRM distributed in a vertical plane (Figure 4). The directions appear to be unrelated to depth, and the plane containing the NRM vectors does not contain the present geomagnetic field, so that

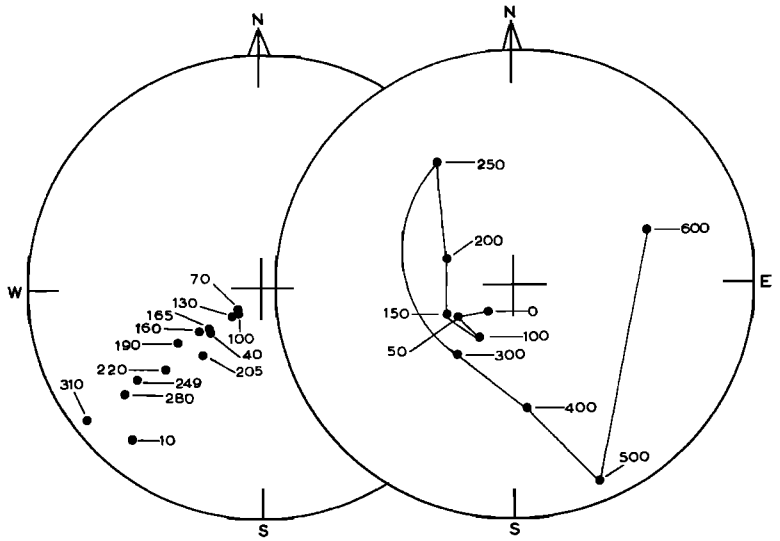


Fig. 4. Demagnetization of granite core, site A9. Left, NRM: numbers give depths of specimens in cm. Right, changes in direction of remanent magnetization for the specimen from 100 cm depth, during demagnetization. The numbers give the maximum demagnetizing field in gauss. All directions north-seeking down.

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viscous magnetization cannot be invoked to explain the distribution. Figure 4*b* shows the behavior of one specimen, that from 100 cm depth, in progressively increased alternating fields. Instability developed at a field of about 100 to 150 gauss.

Gifford [1961] writes as follows concerning the opaque minerals in the Marangudzi rocks:

'*The Gabbroic Rocks.* Titaniferous magnetite is the most abundant opaque mineral in the gabbroic rocks, and occurs as subhedral to anhedral grains. Ex-solution lamellae of ilmenite parallel to (111) are present and also fine lamellae of ulvöspinel parallel to (100). The lamellae of ulvöspinel form a fine network in the magnetite similar to that described by Ramadohr (1953). In some of the specimens the magnetite shows alteration to martite. Small quantities of pyrrhotite, pyrite, and chalcopyrite may be present, but are not found in all specimens.

'*The Acid Rocks.* Titaniferous magnetite and ilmenite are the common opaque minerals in the acid rocks. The ilmenite occurs as exsolution lamellae in the magnetite, in intergrowths with magnetite, and as discrete grains. The titaniferous magnetite is usually the most abundant opaque mineral, but ilmenite predominates in one or two specimens. Ulvöspinel is not found in the titaniferous magnetite occurring in the acid rocks. In all the specimens studied, the magnetite has been partially altered to martite. Sulphides were not found in the acid rocks.

'*The Undersaturated Rocks.* Little opaque material is present in many of the undersaturated rocks, but titaniferous magnetite is again the most common ore mineral. It usually contains exsolution lamellae of ilmenite and may show alteration to martite. Small quantities of pyrite, pyrrhotite, and chalcopyrite are present in some specimens.'

There appears to be no obvious reason for the differences in the remanent magnetizations.

One site (A8, microgabbro) required fields between 400- and 800-gauss maximum amplitude to remove severe lightning effects and then gave well-grouped magnetizations nearly opposed to the directions given by sites A3, 7, 10, and 11. The directions at these four sites are reversed with regard to the present field. The parameters for the natural remanent magnetizations, the magnetizations after treatment in alternating magnetic fields, and the site pole positions are given in Table 3 for the sites A3, 7, 8, 10, and 11. The site mean directions of stable magnetization are shown in Figure 5.

Paleomagnetic results, the Mateke Hills ring complexes. All eight sites gave scattered directions of NRM. Seven sites gave well-grouped and stable directions of magnetization upon partial demagnetization in alternating magnetic fields. The specimens from sites B2, C3, D1, D2, and D3 showed upon alternating-field treatment the classical behavior of a rock which has stable TRM with superimposed IRM of lower coercivity: the direction first moved, in one plane or successively in two planes, as the field was increased to a certain value, and then remained constant for further increase of field. At sites B1 and C1 some care was necessary to find the best demagnetizing field, as some specimens were verging upon instability. At site B1 one core was discarded because specimens cut from it gave discordant directions of magnetization. At site C2 the directions of magnetization of cores remained scattered even after alternating-field treatment, and the results from this site have been discarded.

Six sites (B1, B2, C3, D1, D2, and D3, Table 2) gave closely grouped directions of stable magnetization reversed with respect to the present geomagnetic field and in good agreement

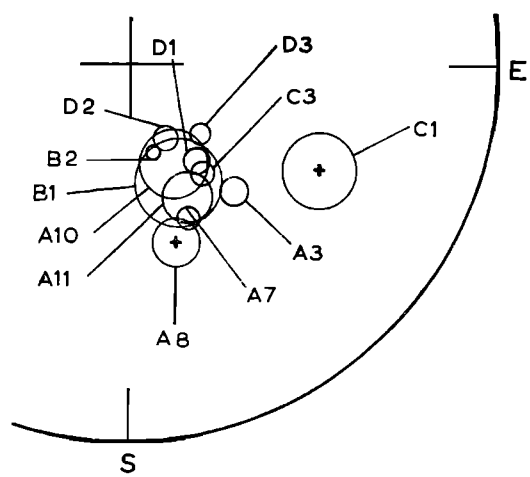


Fig. 5. Circles of 95 per cent confidence for twelve site mean directions of magnetization. Open circles, north-seeking pole down; circled crosses, south-seeking pole down. The direction is in the lower-hemisphere projection.

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TABLE 3. Mean Directions of Magnetization and Site Pole Positions from Twelve Sites

Complex	Site No.	N_c	N	E	M_N D, deg	M_N I, deg	M_N R	M_N k	B_{ac}, gauss	M_{ac} D, deg	M_{ac} I, deg	M_{ac} R	M_{ac} k	α_{95}, deg	Site Pole Lat. S	Site Pole Long. E	Site Pole N or S
Marangudzi	A3	1	11	0	171.0	+7.6	10.939	166	300	139.7	+43.0	10.928	140	3 9	53.1	108 1	N
	A7	1	12	0	(112.5)	(+28.8)	5.214	(1.6)	300	157.6	+43.2	11 949	216	3.0	69.3	107.9	N
	A8	9	18	0	(328.3)	(−45.6)	5.517	(2 3)	400–800	340.0	−38.5	8.802	40.3	8.2	71.4	115.8	S
	A10	9	18	4	(39.7)	(−23 6)	6.190	(2.8)	600–800	155.1	+58 3	8.683	25.2	10.4	62.8	76.3	N
	A11	9	18	0	(238.6)	(−45.0)	2.764	(1.3)	600	156.1	+48.3	8.763	33.7	9.0	67.3	96.9	N
Dembe-Divula	B1	4	8	1	(84.3)	(+28.9)	1.055	(1.0)	150–250	156.4	+53 0	3 933	45.0	13 8	66.1	86.9	N
	B2	1	9	0	(153.2)	(+65.6)	5.464	(2.3)	300	163 8	+62 9	8.989	696	2 0	63.9	58.1	N
Masukwe	C1	8	8	0	(353.7)	(−38.7)	4.912	(2.3)	250	298.8	−29.7	7.773	30.8	10.1	32.1	115.3	S
	C3	7	7	0	(145.8)	(+38.2)	2.418	(1.3)	100	146.2	+52.0	6.984	367	3 2	58.2	94.1	N
Northern	D1	8	8	0	(139.1)	(−11.9)	4.173	(1.8)	150	145 2	+55 5	7.973	261	3.4	56.5	88.1	N
	D2	8	8	0	(162.3)	(+7.3)	5.709	(3 1)	500	152.9	+65 4	7.973	256	3.5	56.2	64.8	N
	D3	8	8	0	(153.5)	(−3.7)	3.210	(1.5)	150	133.9	+60.5	7.977	308	3.2	46.7	83.1	N

N_c = number of cores drilled.
N = number of specimens measured.
E = number of specimens excluded in calculating result.
M_N denotes natural remanent magnetization.
M_{ac} denotes magnetization after treatment in alternating field of maximum amplitude B_{ac}.
D = declination of remanent magnetization.
I = inclination of remanent magnetization.
R = resultant of N_c unit vectors where $N_c \neq 1$;
= resultant of N unit vectors where $N_c = 1$.
$k = (N - 1)/(N - R)$ = estimate of Fisher's κ parameter.
α_{95} = semiangle of cone of 95 per cent confidence for mean direction.
The numbers in parentheses refer to the 'resultants' of data which are random on *Watson*'s [1956*b*] criterion.

with the directions given by sites A3, A7, A10, and A11 at Marangudzi. These six sites are believed to give the directions of thermoremanent magnetizations acquired at the date of intrusion. Site C1 gave a discordant direction of magnetization, which differs by about 35° from the reversed direction with regard to the other sites. The parameters of the mean site directions of magnetization for seven sites in the Mateke Hills complexes are given in Table 3 and are shown in Figure 5.

Age of the Marangudzi complex. When it became clear that a paleomagnetic pole position would be secured from the Marangudzi complex, Dr. J. A. Miller very kindly undertook to date samples from this complex from K-Ar decay. Dr. Miller has supplied the results given in Table 4. The analytical data will be published in full by Miller and Gifford. The excellent consistency of the results leaves little doubt that the age of the Marangudzi complex is close to 190 m.y.

Combination of results from Marangudzi and the Mateke Hills. The Mateke Hills ring complexes are within 80 km of Marangudzi, and complexes *A*, *B*, and *C*, together with the Marumbe granite intrusion marked *E* (Figure 1), lie along a straight line parallel to the Limpopo lineament. Despite important differences in detail, there are general resemblances among ring complexes A, B, C, and D with regard to the range of rock types (gabbro to granite) and over-all shape and dimensions of the complexes. From geological considerations alone it is probable that the intrusion of Marangudzi and the ring complexes at Mateke Hills were approxi-

TABLE 4. Ages of Samples from Marangudzi

Sample	Age, m.y.
Biotite from biotite gabbro	192 ± 10
Biotite from biotite foyaite	182 ± 10
Biotite from malignite	196 ± 10
Biotite from gabbro	187 ± 10
Hornblende contaminated with feldspar, from pulaskite	195 ± 10
Granite (total rock)	186 ± 10

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mately contemporaneous. The paleomagnetic measurements tend to support such a correlation. Ten sites in the four complexes give mean directions of TRM approximately reversed with respect to the present geomagnetic field. It is reasonable to assume that the same period of reversed field is represented in both localities.

On the assumption of a geocentric dipole field the four reversed sites at Marangudzi give for the position of the mean north paleomagnetic pole: latitude 63.7°S, longitude 97.6°E, K = 67, A_{95} = 11.3°, where K is Fisher's dispersion parameter [*Fisher*, 1953] and A_{95} is the radius of 95 per cent confidence for the mean pole position. The six reversed sites from Mateke Hills give a mean north paleomagnetic pole position: latitude 58.6°S, longitude 79.7°E, K = 66, A_{95} = 8.3°. These mean pole positions are 10.1° apart. That they are not significantly different at the 95 per cent confidence level is confirmed by the application of the Snedecor F test [*Watson*, 1956*a*].

The final pole position has been calculated using the combined results from Marangudzi and Mateke Hills.

Field reversal. It is probable that the two 'normal' sites (A8, C1) were magnetized after a reversal of the geomagnetic field relative to its sense at the times of cooling of the ten sites which agree. If the sign is disregarded, the stable component of magnetization at site A8 agrees reasonably well in direction with the ten reversed sites. At site C1, however, the direction of magnetization differs from those at the ten sites by some 35°, after allowance is made for reversal of sense (Figure 5). The field therefore seems to have been considerably disturbed at the time of magnetization of C1. The study by *van Zijl et al.* [1962] suggests that the field can undergo considerable deflections before and after a reversal. It is an interesting possibility that the reversal at Marangudzi is the same as that in the Stormberg lavas.

The 'normal' site at Marangudzi (A8) is in microgabbro, a late intrusion into the main gabbro mass after the latter had cooled. Site A8 therefore acquired normal magnetization after the gabbro sites A7 and A10 had recorded a reversed field. The syenite-foyaite ring was intruded after the gabbro and granite, but the pulaskite, site A3, which is in this ring, shows reverse magnetization. The microgabbro may, therefore, postdate not only the main gabbro but also the granite ring and the syenite-foyaite ring. If the geomagnetic field changed sign it apparently changed from reverse to normal sense near the end of the formation of the Marangudzi intrusion.

Two specimens from each of the 'normal' sites A8 and C1, and one specimen from each of the reversed sites A7 and A10, were heated to 720°C and allowed to cool in the geomagnetic field. All these specimens acquired normal thermoremanent magnetizations. As far as this evidence goes, it suggests that sites A8 and C1 record a geomagnetic field reversal at the end of the intrusive episode rather than that self-reversal is involved.

Pole position. The ten reversed sites listed in Table 3 give, on the assumption of a geocentric dipole field, the mean north paleomagnetic pole position: Latitude 60.9°S, longitude 86.2°E, K = 57, A_{95} = 6.5°. If the two 'normally' magnetized sites A8 and C1 are included, with an assumed reversal, the mean pole position from twelve sites becomes: latitude 60.0°S, longitude 90.2°E, K = 31, A_{95} = 7.9°. The difference between the two estimates is 3.0°, which is not significant. Since the geomagnetic field appears to have been disturbed when one of the two 'normal' sites (C1) was magnetized, the former estimate, from the reversed sites

TABLE 5. Ages of Specimens from the Lupata Volcanics

Sample No.	Mineral	Wt. % K	Av. % K	$^{x}Ar^{40}/K^{40}$	% Air Ar^{40}	Calculated Age, m.y.
GA731 (IV)	Anorthoclase	5.364, 5.387	5.376	6.651×10^{-3}	4.7	110.5
GA732 (VI)	Anorthoclase	4.528, 4.534	4.531	6.412×10^{-3}	1.8	106.7

$\lambda = 5.30 \times 10^{-10}$ yr^{-1} $\quad \lambda e = 0.584 \times 10^{-10}$ yr^{-1}
$K^{40} = 1.19 \times 10^{-2}$ atom % K $\quad ^{x}Ar^{40}$ = radiogenic argon

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alone, is preferred. This choice is perhaps somewhat arbitrary, but it does not significantly affect the result.

Gough and Opdyke [1963] have remarked, in discussing the agreement between the pole positions given by the Karroo basalts and the Lupata alkaline volcanics, that the pole might not have moved greatly relative to Africa during the Jurassic and Cretaceous periods. Dr. I. McDougall has very kindly dated two specimens of anorthoclase from the Lupata volcanics, using K-Ar decay, and has given the results shown in Table 5. Since radiogenic Ar commonly leaks from K feldspars, these results give a minimum age.

With this reservation, it is now possible to extend the remark then made:

Rock Unit	Age, m.y.	Paleomagnetic Pole		
		Lat.	Long.	A_{95}
Marangudzi–Mateke Hills	190	60.9°S	86.2°E	6.5°
Lupata volcanics	109	61.8°S	79.0°E	4.1°

The difference between the mean pole positions relative to Africa is 7.2°; this difference is not significant at the 95 per cent confidence level on Watson's F test [*Watson*, 1956*a*]. The two formations differ in age by about 80 m.y.

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